WEB APPLICATION PROJECT

TRACK YOUR SPENDING

INTRODUCTION AND AIMS TO THE PROJECT

This project will be broken down into **two** parts:

In **part 1A**, you will identify the web application option you would like to develop. You will need to submit a reflection on why you have chosen to develop your identified option, and how you will design your database for the data stored. More information is given below.

In **part 1B**, you will proceed with the development of the web app based on your design with consideration of UI/UX principles.

LEARNING OUTCOMES FOR WEB APP PROJECT

- Apply knowledge of algorithms, relational databases, web development and programming skills to solve a real-world problem.
- Apply usability principles in the design of web applications.
- Create a detailed proposal for your product.
- Demonstrate and apply G.R.A.C.E competencies in the development of your project.
- Apply good time management.

Individual Project. Each student is expected to produce and submit his/her own web application.

You will choose <u>one</u> option out of the seven options below as your project.

Software needed:

- Python IDLE
- Notepad++ or Notepad
- Python Flask, sqlite3
- DB Browser

SUBMISSION DEADLINE:

<u>29 June 2026, Monday</u>

ASSESSED ITEMS

- 1 X Softcopy of **Project Reflection** in a weow (to be submitted online)
- Name your project proposal in the format: `<Class>_<Your Name>.docx`

Any form of plagiarism will be considered as serious violation of academic honesty.

PROJECT BRIEF
OPTION 1
PERSONAL MONEY MANAGEMENT

Many individuals struggle to manage their spending and keep track of their available funds across various accounts. Your team is to design a web-based solution that allows users to log expenses, categorise them, and monitor balances in their bank and cash accounts.

Task 1A – Database Design

Design a fully normalised (3NF) relational database for your application. Your design should support the following:

- Users with individual login credentials (e.g.: email and password)
- Multiple spending accounts per user (e.g.: DBS savings, OCBC 360, cash)
- Expense records including date, amount, category, spending account, and description
- Income records including date, amount, spending account, and description
- Updating of the current balance in each spending account (e.g.: income, expense)
- Categorisation of expenses (e.g.: Food, Transport, Entertainment)

Include in your Project Reflection:

- Table descriptions
- ER diagram
- SQL CREATE TABLE statements

Task 1B – Web Page Design and Development

Create wireframes for the key web pages. Include in your proposal:

- Layouts for user registration, login, and dashboard
- Pages for adding, viewing, editing, and deleting income and expenses
- A basic summary of income and expenses (e.g.: by month, year, and/or expense category).
- Show clear layout, components and data input/ output.

PROJECT BRIEF

OPTION 2
STUDENT STUDY PLANNER

Many students struggle to manage their revision schedule, homework deadlines, and subject priorities during busy school terms. Your task is to design a web-based study planner that allows students to organise their tasks, track their progress, and monitor upcoming deadlines.

Task 1A – Database Design

- Design a fully normalised relational database in 3NF for your application. Your design should support the following:
- Users with individual login credentials, such as email and password
- Subjects created by each user, such as Computing, Mathematics, Economics, or Chemistry
- Study tasks including task title, description, subject, due date, priority level, and completion status
- Study sessions including date, start time, end time, subject, and remarks
- Categorisation of tasks, such as Homework, Revision, Project, Consultation, or Test Preparation
- Updating of task status, such as Not Started, In Progress, or Completed

Include in your Project Reflection:

- Table descriptions
- ER diagram
- SQL CREATE TABLE statements

Task 1B – Web Page Design and Development:

- Pager for user registration, login and a dashboard when logged in
- Pages for adding, viewing, editing, and deleting study tasks
- Pages for recording and viewing study sessions
- A basic summary of study progress, such as number of completed tasks by subject, upcoming deadlines, or total study hours by week

PROJECT BRIEF
OPTION 3
PERSONAL FITNESS AND HEALTH TRACKER

Many individuals want to build healthier habits but find it difficult to keep track of their exercise routines, body measurements, and progress over time. Your task is to design a web-based fitness tracker that allows users to record their workouts and monitor their personal fitness goals.

Task 1A – Database Design

Design a fully normalised relational database in 3NF for your application. Your design should support the following:

- Users with individual login credentials, such as email and password
- Fitness goals created by each user, such as weight loss, strength training, stamina improvement, or general fitness
- Workout records including date, workout type, duration, intensity, and remarks
- Exercise records including exercise name, number of sets, repetitions, and/or distance
- Body measurement records including date, weight, height, and optional notes
- Categorisation of workout types, such as Cardio, Strength, Flexibility, Sports, or Others

Include in your Project Reflection:

- Table descriptions
- ER diagram
- SQL CREATE TABLE statements

Task 1B – Web Page Design and Development:

- Pages for user registration, login, and dashboard
- Pages for adding, viewing, editing, and deleting workout records
- Pages for recording and viewing body measurements
- A basic summary of fitness progress, such as total workout duration by week, workout frequency by category, or progress towards fitness goal.
- Show clear layout, components and data input/output.

PROJECT BRIEF
OPTION 4
BOOK READING LOG

Many students and readers wish to develop a consistent reading habit but may not have a structured way to track books they are reading, books they have completed, and personal reflections. Your task is to design a web-based reading log that allows users to record books, track reading progress, and review their reading habits.

Task 1A – Database Design

Design a fully normalised relational database in 3NF for your application. Your design should support the following:

- Users with individual login credentials, such as email and password
- Book records including title, author, genre, total number of pages, and publication year
- Reading records including start date, completion date, current page, and reading status
- Personal reviews including rating, reflection, and review date
- Categorisation of books by genre, such as Fiction, Non-Fiction, Mystery, Science, Fantasy, or Biography
- Updating of reading status, such as Want to Read, Currently Reading, Completed, or Dropped

Include in your Project Reflection:

- Table descriptions
- ER diagram
- SQL CREATE TABLE statements

Task 1B – Web Page Design and Development:

- Pages for user registration, login, and dashboard
- Pages for adding, viewing, editing, and deleting book records
- Pages for updating reading progress and adding reviews
- A basic summary of reading progress, such as number of books completed, pages read by month, or books by genre
- Show clear layout, components and data input/output. Explain how usability principles are applied in your design.

PROJECT BRIEF

OPTION 5
EVENT SIGN-UP AND ATTENDANCE SYSTEM

Schools often organise workshops, enrichment programmes, talks, and CCA activities. It can be difficult to manage event sign-ups, participant details, and attendance records manually. Your task is to design a web-based event sign-up system that allows users to register for events and allows organisers to monitor sign-ups and attendance.

Task 1A – Database Design

- Design a fully normalised relational database in 3NF for your application. Your design should support the following:
- Users with individual login credentials, such as email and password
- Events including event title, description, date, time, venue, and maximum capacity
- Event categories, such as Workshop, Talk, Competition, CCA Activity, or Enrichment
- Sign-up records showing which users have registered for which events
- Attendance records indicating whether a registered participant was present, absent, or excused
- Updating of sign-up status, such as Registered, Withdrawn, Attended, or Absent

Include in your Project Reflection:

- Table descriptions
- ER diagram
- SQL CREATE TABLE statements

Your web application should have the following:

- Pages for user registration, login, and dashboard
- Pages for adding, viewing, editing, and deleting event records
- Pages for users to sign up for or withdraw from events
- Pages for organisers to mark attendance
- A basic summary of event participation, such as number of sign-ups per event, attendance rate, or events by category
- Show clear layout, components and data input/output.

PROJECT BRIEF
OPTION 6
ONLINE SHOPPING / POINT-OF-SALES SYSTEM

Many small businesses, school-based fundraisers, and student-run stalls need a simple system to manage products, process customer purchases, and keep track of sales. Your task is to design a web-based e-shop or point-of-sales system that allows users to browse products, place orders, and view sales records.

Task 1A – Database Design

- Design a fully normalised relational database in 3NF for your application. Your design should support the following:
- Users with individual login credentials, such as email and password
- Product records including product name, description, price, stock quantity, and product category
- Product categories, such as Food, Drinks, Stationery, Electronics, Books, or Others
- Customer orders including order date, order time, order status, and total amount
- Order items showing the products purchased in each order, quantity bought, and subtotal
- Updating of product stock quantity after an order is placed
- Optional payment status, such as Pending, Paid, Cancelled, or Refunded

Include in your proposal:

- Table descriptions
- ER diagram
- SQL CREATE TABLE statements

Task 1B – Web Page Design and Development

- Pages for user registration, login, and dashboard
- A product catalogue page for users to browse available products
- Pages for adding, viewing, editing, and deleting product records
- A shopping cart or order creation page
- A checkout or order confirmation page
- Pages for viewing past orders and order details

- A basic sales summary, such as total sales by day, month, product category, or best-selling products
- Show clear layout, components and data input/output.